

# **Safety Gear Monitoring System Project Report**

## **A Real-Time Computer Vision Application for Workplace Safety**

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## **1. Introduction**

The Safety Gear Monitoring System is an advanced real-time computer vision application designed to enhance workplace safety by ensuring compliance with safety gear requirements, such as vests and helmets. Developed by a dedicated team, the system integrates the YOLOv8 object detection model, DeepSort tracking algorithm, a PyQt5-based graphical user interface (GUI), and serial communication for hardware control. It detects and tracks safety gear in video feeds, provides visual and audible alerts for non-compliance via LEDs and a buzzer, and maintains hourly video backups for post-incident analysis. This solution is tailored for high-risk environments like construction sites and manufacturing facilities, promoting safety and accountability.